Membership inference under a version-space training oracle

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Setting

Let $\mathcal{X}$ be an input set, $\mathcal{H} \subseteq \{0,1\}^{\mathcal{X}}$ a finite hypothesis class, and $c \in \mathcal{H}$ a target (the realizable case). A labelled point is $z = (x, c(x))$. For a finite set S of labelled points the *version space* is

$$V(S) = \{h \in \mathcal{H} : h(x) = y \text{ for every } (x,y) \in S\}.$$

The oracle. On training set S the learner outputs $h \sim \mathrm{Unif}(V(S))$.

The utility condition. Every output fits the training set, so training accuracy is 1. This is the conditioner on success: a learner that outputs a constant is excluded, because a constant is not in $V(S)$ unless the target is constant on S. Test accuracy is not fixed by the oracle, and the result below says it is exactly what governs membership leakage.

Relation to SDSOJ19 (proven, elementary). With 0-1 loss and a uniform prior on $\mathcal{H}$, the Gibbs posterior $p(h \mid S) \propto \exp\big(-\frac{1}{T}\sum_{z\in S}\ell(h,z)\big)$ converges as $T \to 0$ to $\mathrm{Unif}(V(S))$ when $V(S) \neq \emptyset$. So this oracle is the zero-temperature case of their model, not a new one.

The game

The adversary knows $\mathcal{H}$, c's labels on all points it is given, and the rest of the training set T, a set of $n-1$ labelled points. This is the strong "other memberships known" adversary of SDSOJ19.

1. Draw z_0, z_1 i.i.d. from the data distribution and $b \in \{0,1\}$ uniformly.
2. Train on $S = T \cup \{z_b\}$, giving $h \sim \mathrm{Unif}(V(S))$.
3. The adversary receives T, z_0, z_1 and access to h, and outputs b'.

White-box access is h itself. Black-box access is queries $x \mapsto h(x)$.

Write $V_i = V(T \cup \{z_i\})$, $V_{01} = V_0 \cap V_1$, and $r_i = |V_{01}|/|V_i|$. The ratio r_i is the probability that a model trained with z_i labels the other challenge point correctly, so it is test accuracy on a fresh point.

Claim

Proposition. The Bayes-optimal adversary's decision depends on h only through the two values $h(x_0)$ and $h(x_1)$. Its success probability is

$$\Pr[b' = b] = 1 - \tfrac{1}{2}\min(r_0, r_1),$$

so its advantage $2\Pr[b' = b] - 1$ equals $1 - \min(r_0, r_1)$.

Proof. Given b, $p(h \mid b) = \mathbf{1}[h \in V_b]/|V_b|$. Every h that can occur lies in $V(T)$, so $h \in V_b$ holds exactly when $h(x_b) = c(x_b)$. The likelihood ratio is therefore a function of $f_i = \mathbf{1}[h(x_i) = c(x_i)]$, $i = 0, 1$, and of the constants $|V_0|, |V_1|$, which do not depend on h. The four cases:

- $(f_0, f_1) = (1, 0)$ occurs only under $b = 0$, with probability $1 - r_0$. Guess 0.

- $(0,1)$ occurs only under $b=1$, with probability $1-r_1$. Guess 1.
- $(1,1)$ occurs under b with probability r_b. Guess the b with the larger r_b, the challenge point that leaves fewer consistent hypotheses.
- $(0,0)$ never occurs.

Summing, $\Pr[b'=b] = \frac{1}{2}\left[(1-r_0)+(1-r_1)+\max(r_0,r_1)\right] = 1-\frac{1}{2}\min(r_0,r_1)$. $\square$

What it says

1. **Utility sets leakage.** If a model trained without a point already labels it correctly with probability 1, then $r=1$ and the advantage is 0. Leakage is exactly the failure to generalize to the challenge point. This is the overfitting link of Yeom et al. (CSF 2018) and the long-tail picture of Feldman (STOC 2020), in one line, for this oracle.
2. **No black-box against white-box gap.** Two black-box queries achieve the white-box optimum. So under this oracle, and under the Gibbs oracle of SDSOJ19, the gap the programme looks for cannot appear. A gap needs a learner that leaves the Gibbs family: weights that carry more than the function, or a choice within the version space that depends on S beyond consistency, as an implicit bias of SGD might.
3. **Where computation enters.** The only step that needs computation is the $(1,1)$ case, comparing $|V_0|$ and $|V_1|$, a counting problem over the class. An adversary that guesses there succeeds with probability $1-\frac{1}{4}(r_0+r_1)$, so it loses $\frac{1}{4}|r_0-r_1|$. Whether that counting is hard for natural classes is open here. Counting consistent hypotheses is hard for rich classes in general, which is a heuristic expectation and not proven for any class in this note.